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REVIEW ARTICLE

# Unraveling Etiological Indications and Therapeutic Implications of Familial Cerebral Cavernous Malformations in the Dawn of Gene Therapy for Monogenic Conditions

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**Abstract:** Cerebral Cavernous Malformations (CCMs) are vascular anomalies in the central nervous system that arise from both genetic and non-genetic factors, and can cause hemorrhage, seizures, and neurological deficits. Approximately 80% of CCMs are sporadic, while 20% are Familial (FCCMs), an autosomal dominant, monogenic disorder characterized by multiple lesions and severe clinical manifestations. Over the past three decades, linkage analyses have identified *KRIT1/CCM1*, *MGC4607/CCM2*, and *PDCD10/CCM3* as major pathogenic genes in FCCMs. However, existing surgical and pharmacological treatments have not adequately prevented disease progression, underscoring the need for more effective strategies. Recent advancements in gene editing tools and delivery systems have transformed gene therapy from a laboratory concept to a clinical reality, offering renewed hope for FCCM patients. Given the multifactorial nature, complexity, and neurological comorbidities of FCCMs, exploring non-surgical gene therapies provides a promising approach for addressing these cerebrovascular lesions. This review summarizes the latest progress in gene editing for FCCMs and examines its therapeutic potential, while acknowledging both the promising benefits and the remaining uncertainties in this evolving field.

**Keywords:** Cerebral cavernous malformations, gene therapy, monogenic disease, CRISPR Cas9, base editing, central nervous system.

## 1. INTRODUCTION

Cerebral Cavernous Malformations (CCMs), also known as cavernomas, are characterized by multiple lesions in the Central Nervous System (CNS), consisting of collections of structurally abnormal, slow-flowing capillaries [1, 2]. These lesions are more commonly found in the supratentorial region than in the infratentorial region [2]. The majority of CCM cases, referred to as sporadic CCMs, involve a single lesion and are typically asymptomatic and non-hereditary. In contrast, Familial Cerebral Cavernous Malformations (FCCMs) are a monogenic disorder characterized by multiple lesions and more severe clinical symptoms. The population incidence of FCCMs is estimated to be between 1 in 3,300 and 1 in 10,000 [3]. Affected parents have a 50% chance of passing the genetic mutation to their offspring, leading to significant familial clustering of FCCMs. Consequently, each generation within an affected family has a high probability of carrying pathogenic genes [4]. Despite this classic autosomal dominant inheritance pattern, some studies hint at the involvement of secondary genetic or epigenetic modifiers that may influence the penetrance and severity of FCCMs. Future work dissecting such modifier loci could be pivotal in understanding cases where the known causal genes do not fully explain the clinical outcome.

## 2. METHODOLOGY

This narrative review summarizes existing research on Familial Cerebral Cavernous Malformations (FCCM) and gene therapy without employing systematic review or meta-analysis protocols. Literature was retrieved from PubMed and Web of Science up to 2025 using a combination of MeSH and free-text terms, including “FCCM,” “Gene Therapy,” “CRISPR,” “Base Editing,” and “Prime Editing,” with Boolean operators (AND/OR) to structure the search. A snowballing method was also used to identify additional

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relevant studies from reference lists. Included studies focused on FCCM pathogenesis, gene editing tools (*e.g.*, CRISPR-Cas9, base editing, prime editing), delivery systems (*e.g.*, AAV, LNP), and preclinical or translational research. Priority was given to recent peer-reviewed publications from the past five years, with foundational studies on KRIT1, CCM2, and PDCD10 genes included where relevant. Exclusion criteria comprised studies lacking relevance, non-English publications, duplicates, and those without primary data. The literature was organized according to the sequence: FCCM pathogenesis, gene editing technologies, delivery systems, and clinical considerations. Where findings were inconsistent, comparisons were made based on methodological differences, sample sizes, and experimental models. The review emphasizes key challenges in FCCM gene therapy—such as blood-brain barrier delivery, correction thresholds for mosaic lesions, and immunogenicity—and discusses the potential applicability of newer gene editing tools like glycosylase base editing and RNA editors, such as Cas13a, in addressing these issues.

## 3. PATHOGENIC GENES OF FCCMs

Familial Cerebral Cavernous Malformations (FCCMs) are recognized as an autosomal dominant monogenic genetic disorder, which follow a two-hit mutation model: germline mutation plus a secondary somatic event (Fig. **1**). Through linkage analysis, pathogenic genes associated with FCCMs, including *KRIT1/CCM1*, *MGC4607/CCM2*, and *PDCD10/CCM3*, have been identified [5, 6]. However, these three mutations are not found in more than 5% of FCCM cases [7]. Global gene sequencing studies have identified over one hundred mutation sites within these three chromosomal regions. Significant genetic heterogeneity is observed in the mutation sites across different populations (such as Hispanic, Latino, and Mexican populations), and notable differences exist in the proportion of FCCMs caused by *KRIT1*, *MGC4607*, and *PDCD10* mutations [8]. In the Han population, nearly ten mutation sites have been identified in FCCM families, with the vast majority located within the *KRIT1* gene [9-15]. Mutations in CCM genes lead to loss of protein function, resulting in cellular dysfunction and disease. The most common mutation type is protein truncation, although missense mutations, leading to protein misfolding, are also rare [6]. Notably, a residual subset of FCCM patients (estimated to be 5% or more) remains genetically unexplained, suggesting that large deletions, cryptic regulatory variants, or even newly discovered genes might be implicated. Establishing uniform standards for variant interpretation—*via* functional assays or advanced long-read sequencing—will be essential in narrowing this gap.

Extensive laboratory research spanning 20 years has explored the molecular mechanisms underlying CCM development [16, 17]. The three genes, *KRIT1*, *CCM2*, and *PDCD10*, collectively regulate cell proliferation, network formation, and endothelial growth within a larger signaling pathway. These genes directly interact as part of a heterotrimeric adaptor complex known as the CCM complex, which explains how mutations in three distinct genes can result in the same disease while encoding three non-homologous proteins [18, 19]. Mutations in the CCM genes are believed to impair critical surface interactions between these proteins and numerous other binding partners [6]. The loss of the CCM complex leads to the upregulation of MEKK3 signaling, which subsequently triggers the pathological overexpression of downstream transcription factors, namely KLF2 and KLF4, along with an increase in Rho-associated protein kinase (ROCK) activity. These molecular events contribute to the development of endothelial cell pathology [20]. Notwithstanding the prominence of the MEKK3–KLF2/4 pathway, additional overlaps with the PI3K-mTOR or Wnt–β-catenin cascades may exist, and this cross-talk could be particularly relevant when exploring combination therapies or understanding different clinical courses in patients.

## 4. CLINICAL MANIFESTATIONS AND CELLULAR PATHOLOGY

CCMs can occur at various locations within the CNS, with supratentorial lesions being more prevalent than infratentorial lesions [2]. Pathologically, FCCM lesions do not show significant differences from sporadic CCM. However, familial cases are more likely to affect multiple organs, particularly the brain, retina, skin, vertebrae, liver, and kidneys [21]. Patients with FCCM may experience symptoms such as seizures (40%-70%), focal neurological deficits (25%-50%) without intracranial bleeding, nonspecific headaches (10%-30%), and intracranial hemorrhage (25%-32%), which may be confined to the lesion or extend beyond it. Hemosiderin deposits surrounding individual cavernomas indicate remote leakage caused by abnormal capillaries [7, 22, 23]. Early detection of FCCM is challenging, with a significant proportion of cases (20%-50%) remaining asymptomatic and often being incidentally discovered during head imaging [24]. Nevertheless, as lesions evolve, recurrent microhemorrhages may gradually accumulate iron deposits, potentiating both local neurotoxic effects and inflammation. Most non-bleeding lesions are asymptomatic, resulting in a low clinical diagnosis rate. However, when bleeding occurs within a lesion, it can manifest as symptoms such as headaches, seizures, stroke, or neurological dysfunction. Lesions located in the brainstem or critical functional areas can lead to paralysis, aphasia, respiratory disorders, and even death, posing a serious threat to the life and well-being of affected individuals [25]. Significantly, the mosaic structure of CCMs—where not all endothelial cells harbor the primary mutation—adds further complexity to predicting which lesions will become symptomatic. FCCMs are characterized by multiple lesions, an earlier age of onset in children, and relatively severe symptoms. CCMs are the second most common vascular malformation of the CNS, after developmental venous anomalies [2]. Recent studies report a prevalence of CCM ranging from 0.1% to 0.8% [26, 27]. The population-based Scottish Intracranial Vascular Malformation Study (SIVMS) estimated an annual incidence rate of 0.56 per 100,000 individuals aged 16 years and older [28]. Currently, most treatment strategies for CCM focus primarily on the sporadic form, aiming to stabilize the lesion and prevent its growth and bleeding [22]. Unfortunately, a significant proportion of patients still face the challenge of experiencing at least one stroke event during their lifetime [21]. With advancements in neuroimaging technology, it is now possible to accurately assess the number, location, and

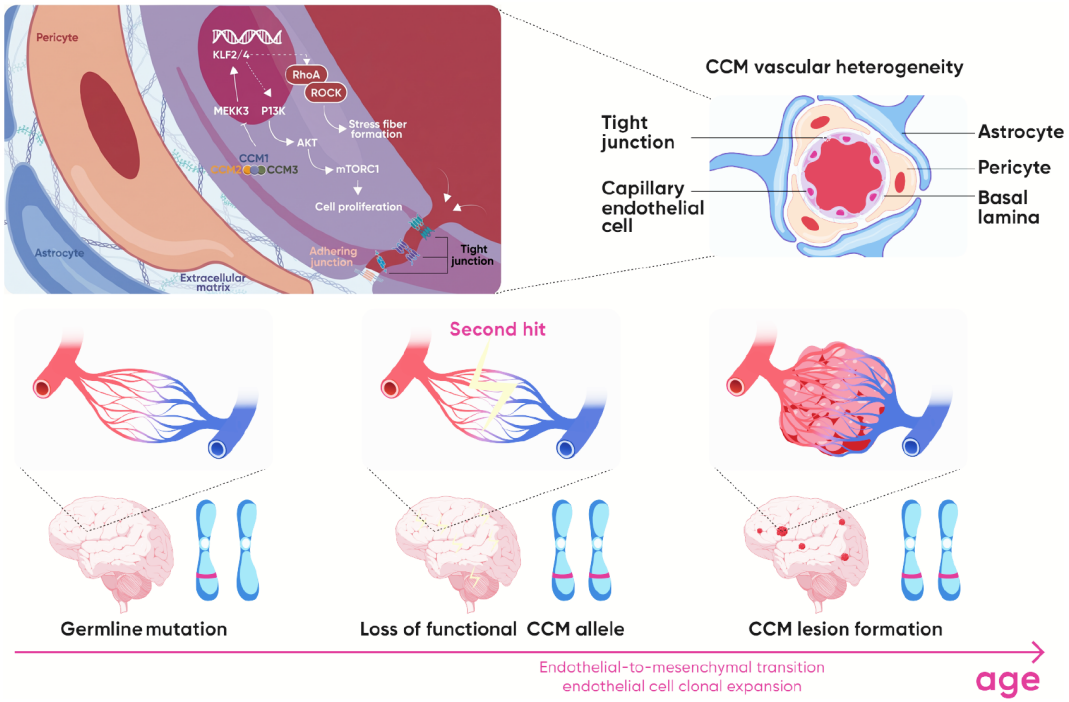


**Fig. (1).** Pathogenic Mechanism of FCCMs. The pathogenesis of FCCMs arises from a two-hit mutation mechanism. The first hit is a germline mutation, while the second hit is a somatic mutation, leading to multifocal lesions. Mutations in CCM genes result in impaired protein expression and function, disrupted cell-cell connections, and increased cellular proliferation, ultimately causing vascular malformations and rupture. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

bleeding stage of CCM lesions using gradient echo sequences or Magnetic Resonance Imaging (MRI). However, the unpredictable nature of bleeding and the lack of effective preventive measures indicate that FCCM patients require lifelong monitoring for the risk of bleeding in their brain lesions.

The exploration of new therapies has faced significant challenges. Clinical trials conducted by a joint team from the University of Chicago and Johns Hopkins University investigated the role of Atorvastatin, a RhoA/ROCK inhibitor, in controlling the progression and bleeding of CCM lesions. However, no effective evidence has been demonstrated so far [29, 30]. In 2021, a collaborative study between the United States and Germany, published in Nature, suggested that CCM development is driven by the functional deletion of the *KRIT1* gene and the activation of the PI3K-mTOR signaling pathway. The study also proposed that Sirolimus, a specific inhibitor of mTOR, could inhibit the progression of CCM lesions. However, clinical trials in this area have not produced ideal results. Researchers at the University of California discovered that Propranolol, a β1 adrenergic receptor antagonist, could improve CCM lesions. Building on this finding, an Italian research team proposed a clinical trial program called "Treat_CCM," but the latest results published in 2023 were unsuccessful [3, 31, 32]. Such setbacks show the complexity of targeting downstream pathways in CCM biology. Even if a candidate drug shows mechanistic promise in preclinical models, inter-patient heterogeneity and the mosaic nature of CCM lesions can complicate real-world efficacy. Given the current treatment strategies and ongoing research, gene editing tools still hold promise for the treatment and potential cure of FCCM, offering a glimmer of hope for alleviating symptoms.

## 5. GENE EDITING AS A THERAPEUTIC STRATEGY

Familial Cerebral Cavernous Malformation (FCCM) is a monogenic, loss-of-function disorder, making it an ideal target for precision gene repair rather than gene knockout. Contemporary editing platforms fall into two functional classes: disruptive editors, which silence genes, and corrective editors, which restore wild-type sequence. Because FCCM pathogenesis arises from truncated or misspliced CCM proteins, corrective tools—especially Base Editing (BE) and Prime Editing (PE)—are preferred; they can repair single-nucleotide errors without generating double-strand breaks or disruptive indels.

The most frequently mutated locus, *KRIT1*, spans 19 exons; exon 16 encodes Ankyrin repeats, an NPXY motif, and a FERM domain that together anchor RAP1A and ICAP1α, preserving endothelial junctions [33-36]. Similar loss-of-function variants in *CCM2* and *PDCD10* converge on the same pathway. Importantly, FCCM lesions are mosaic, meaning that only a proportion of endothelial cells carry the germline + somatic "second hit." Modelling studies and preclinical data indicate that correcting as little as 15–30 % of these cells can re-establish barrier integrity, dampen ROCK signalling, and halt lesion enlargement. Thus, even partial, high-fidelity repair by BE or PE may deliver clinical benefit while minimizing vector load and off-target risk.

Long-term efficacy depends on whether the chosen platform confers a permanent (DNA-level) or transient (RNA-level) correction. BE and PE write permanent changes and

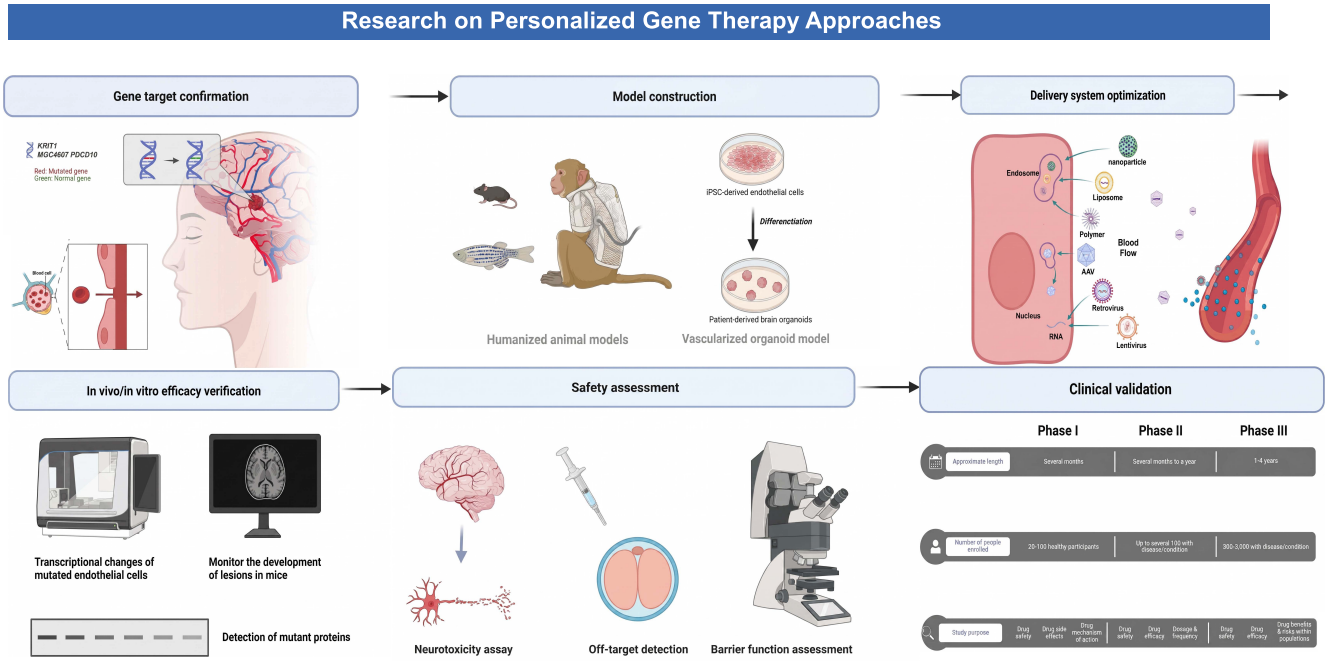


**Fig. (2).** Stepwise Development Pipeline for Personalized Gene Therapies Targeting FCCMs. The process begins with gene target confirmation, focusing on pathogenic variants in KRIT1, CCM2, and PDCD10. Disease models are then constructed using humanized animal models and vascularized brain organoids derived from patient-specific iPSC-endothelial cells. Delivery system optimization involves evaluating various vectors, including nanoparticles, liposomes, AAV, and lentivirus, for CNS targeting and efficient intracellular uptake. Efficacy verification is conducted through transcriptomic profiling, protein detection, and lesion monitoring in preclinical models. Safety assessments include off-target analysis, barrier integrity evaluation, and neurotoxicity testing. The final stage—clinical validation—progresses through Phase I–III trials, which assess safety, dose, and therapeutic benefit across diverse patient cohorts. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

therefore suit one-time or limited-dosing strategies; RNA-guided editors, such as Cas13-ADAR, act only on transcripts and will require redosing, but carry a lower risk of irreversible off-target effects. The sections below compare these options in detail.

### 5.1. *In Vivo vs. Ex Vivo* Editing

Developing FCCM therapies begins with determining whether corrections are made outside the body (*ex vivo*) or directly in the patient (*in vivo*) (Fig. **2**).

*Ex vivo* editing—successfully applied in blood disorders and lymphomas [37-39] —relies on harvesting cells, editing them, and reinfusing them. That workflow is impractical for FCCM: endothelial cells would need to be isolated from multiple CNS lesions, expanded, and surgically re-implanted, all with high immunologic risk [40, 41].

*In vivo* editing delivers CRISPR-based tools (or mRNA/RNP equivalents) straight to the cerebrovasculature [42]. It has already demonstrated clinical benefits in neuromuscular and retinal diseases and, more recently, in a one-shot lipid-nanoparticle therapy for sickle-cell disease [43-45]. For FCCM, the same concept must be adapted to cross the Blood-Brain Barrier (BBB), reach distributed lesions, and limit exposure of healthy vasculature. Permanent editors, such as BE or PE, are attractive here: if even 15-30 % of mutant endothelial cells are corrected, junctional integrity can be restored and lesion growth can be arrested, reducing the need for repeat dosing. By contrast, RNA editors offer transient rescue but would require periodic redelivery—an important trade-off when planning lifelong management.

*In vivo* work is technically demanding: vectors must evade immunity, scale to multilesion dosing, and remain effective if neutralizing antibodies emerge with redosing [46]. Nanoparticles or virus-like particles—possibly combined with transient immunosuppression—are under active investigation to meet these criteria. Continued optimization will determine whether durable, one-time BE/PE therapy or repeatable, transcript-level RNA editing (with its inherently lower permanent off-target risk) becomes the frontline strategy for FCCM.

### 5.2. Delivery Strategies

For *in vivo* therapies to succeed in FCCM, delivery vectors must be stable in circulation, resist immune clearance, cross the BBB, and efficiently target affected endothelial cells [40]. This is especially challenging given the architecture of FCCM lesions—clusters of distended, thin-walled capillaries devoid of normal tissue support [2].

Among delivery vehicles, Adeno-Associated Viruses (AAVs) and Lipid Nanoparticles (LNPs) are the most studied. AAV-BI30 has shown strong tropism for CNS endothelium at low doses in mice, making it a leading candidate for CNS-targeted gene editing. However, AAVs have drawbacks, including immunogenicity, limitations in cargo size, and the challenge of redosing due to neutralizing antibodies [47].

Non-viral systems, particularly lipid- or polymer-based nanoparticles, are emerging as attractive alternatives. These carriers are tunable, low in immunogenicity, and capable of encapsulating large editing complexes [48]. Some preclinical work suggests that transient induction of Endothelial-to-Mesenchymal Transition (EMT) may increase nanoparticle uptake, though safety concerns remain regarding EMT induction in the CNS.

Ultimately, carrier systems must achieve sufficient transduction coverage across dispersed lesions while maintaining high selectivity for mutant endothelial cells. Strategies may include ligand-directed nanoparticles or engineered AAVs with improved endothelial tropism. In FCCM, even a small margin of off-target uptake may carry a serious risk, so precise targeting is critical (Table **1**).

## 6. GENE EDITING TOOLS

Gene editing holds promise for correcting the diverse mutations implicated in FCCM, including transition and transversion point mutations, small deletions, and indels. Transition mutations are implicated in disorders like progeria; transversions underlie diseases like sickle cell anemia; and insertions or deletions are the basis of conditions such as Tay-Sachs and cystic fibrosis. In FCCM, the predominant mutation types are loss-of-function variants, particularly protein-truncating mutations in *KRIT1*, *CCM2*, and *PDCD10*. The heterogeneity of FCCM lesions—including their mosaic structure and variable penetrance—places a premium on tools that combine precision, flexibility, and delivery compatibility with the CNS. A comparative overview of major editing tools is provided in Fig. (**3**) and further detailed in Table **2**, summarizing their mutation scope, durability, feasibility of CNS delivery, and suitability for mosaic FCCM lesions.

Tools must also address an important pathophysiological constraint: in FCCM, correcting only a subset of cells within a lesion may be sufficient to stabilize local vasculature. Thus, tools that achieve partial correction with high fidelity can be therapeutically effective, even without universal correction.

Additionally, the permanence of correction varies depending on the tool. DNA-targeting systems, such as CRISPR-Cas9, Base Editors (BEs), and Prime Editors (PEs), induce lasting genomic changes but carry risks of off-target DNA alterations, especially concerning non-regenerating CNS tissue. In contrast, RNA editing systems, including Cas13-ADAR platforms, offer transient, reversible correction of mutant transcripts—safer for sensitive tissues but requiring repeated delivery [49].

### 6.1. CRISPR-Based Platforms

CRISPR-Cas systems have revolutionized gene editing by enabling the precise and programmable manipulation of genetic material. For FCCM, the choice of the CRISPR tool must consider mutation type, editing precision, delivery compatibility with the CNS, and risks associated with off-target effects.

The most widely used system, Cas9, introduces Double-Strand Breaks (DSBs) at specific DNA loci *via* its RuvC and HNH nuclease domains [50-52]. Repair typically occurs through Non-Homologous End Joining (NHEJ), which is fast but error-prone, or Homology-Directed Repair (HDR), which is more precise but less efficient in post-mitotic cells like brain endothelium [53-55].

To improve specificity and reduce genotoxicity, modified enzymes like Cas9 nickase (Cas9n) and Cas12a/b have been developed. These generate single-strand nicks or staggered ends, potentially minimizing off-target activity. Cas12a also recognizes T-rich PAM sequences, broadening the genome's editable regions. Meanwhile, Cas13 enzymes offer RNA-targeting activity, opening the door for transient transcript-level modulation rather than permanent edits [56, 57].

Despite its utility, the risk of large deletions or chromosomal rearrangements from DSBs remains a serious limitation for CNS applications. Thus, for FCCM, especially in mosaic lesions located in critical brain regions, tools that avoid DSBs—like base editors or prime editors—are increasingly favored. These newer tools offer more control over editing outcomes while still enabling permanent correction, a key therapeutic advantage for inherited cerebrovascular diseases.

### 6.2. Base Editing

Base editing enables precise, single-nucleotide changes without creating DSBs, making it ideal for treating monogenic diseases like FCCM. It relies on a fusion of a catalytically inactive or nickase Cas9 (dCas9/Cas9n) with a cytidine or adenine deaminase, which converts specific bases within a targeted window [58, 59].

Cytidine Base Editors (CBEs) convert C→T (or G→A), while adenine base editors (ABEs) mediate A→G (or T→C). These transitions cover a substantial fraction of known FCCM mutations across *KRIT1*, *CCM2*, and *PDCD10*. Advanced versions, such as BE4max and ABE8e, offer enhanced fidelity, broader PAM compatibility, and reduced off-target rates [58, 60-63].

For FCCM, BE offers several advantages: (i) it can correct pathogenic mutations at the DNA level, offering permanent restoration of gene function; (ii) It is non-disruptive, avoiding the complications of DSBs; and crucially, (iii) it allows correction of mosaic lesions, where editing even a subset of endothelial cells may be sufficient to restore local vascular integrity and suppress inflammatory pathways.

However, base editing has limitations. The editing window is narrow and sequence-and context-dependent. There is a risk of bystander edits at adjacent cytidines or adenines, which must be carefully evaluated, especially in non-regenerating tissues like the brain. Moreover, while BE is generally more efficient than PE, it is limited to transition mutations and cannot address indels or transversions [64].

Nonetheless, for FCCM patients with amenable point mutations, BE offers a compelling balance of efficiency, safety, and therapeutic durability.

**Table 1. Gene delivery vectors targeting endothelial cells for therapeutic genome editing.**

| Vector | Mod / Tropism Tag | Cargo (Gene / editor) | Primary Target | Intended Indication | References |
|---|---|---|---|---|---|
| **VIRAL VECTORS** | | | | | |
| AAV-BI-hTFR1 | 7-aa VP1 loop (transferrin-binding) | *NLS-mScarlet; luciferase* | Brain ECs (hTfR) | CNS disorders | [94] |
| AAV-BR1 | Brain-EC-tropic capsid | *HEXA; HEXB* | Cerebral ECs | Tay-Sachs / Sandhoff | [95, 96] |
| AAV2-GEC | QVLVYRE glomerular peptide | *IdeS* | Glomerular ECs | Inflammatory nephritis | [97] |
| AAV-GDF11 | — | *GDF11* | ECs | Atherosclerosis | [98] |
| AAV-PHP.eB-NeuroD1 | — | *NeuroD1* | ECs | Ischaemic stroke | [99] |
| AAV9-sFLT1 | Soluble FLT1 ectodomain | — | Vascular ECs | Brain AVM | [100] |
| AAV-BI30 | NNSTRGG brain-EC peptide | *NLS-GFP; luc* | Brain microvascular ECs | Endothelial dysfunction | [47] |
| AAV-PHP.V1 | TALKPFL brain-EC peptide | *Cre* | Brain microvascular ECs | — | [101] |
| AAV2/9-NDVRAVS | NDVRAVS EC-binding peptide | *eGFP* | ECs (HSPG) | — | [102] |
| rAAV2-ESGHGYF | ESGHGYF lung-EC peptide | *Luc; GFP* | Pulmonary ECs | — | [103] |
| rAAV2/2[QuadYF-TV]-EC5 | Heptameric EC motif | *Mutant rep-cap* | Retinal ECs | Diabetic retinopathy / AMD | [104] |
| Hybrid AAV (ETP) | Endothelium-targeting peptide | *VWF* | HUVECs | von Willebrand disease | [105] |
| AAV1-pICAM2-SpCas9 | ICAM2 promoter restricts Cas9 | *SpCas9* | Retinal ECs | Retinopathy | [106] |
| Oncolytic AdV | CD40L trimer | *CD40L-IZ* | HUVECs (CD40) | Pancreatic cancer vessels | [107] |
| AdV-VEGF-DΔNΔC | Truncated VEGF-D insert | *VEGF-DΔNΔC* | HUVECs (VEGFR2/3) | Placental insufficiency | [108] |
| rFVIII-LV | — | *ABE / PE tools* | BOECs | Haemophilia A | [109] |
| VE-cadherin LV | VE-cadherin promoter | *VEGFR2 shRNA* | Retinal ECs | Retinopathy | [110] |
| GP64-LV | Bac-GP64 + CD47 cloak | *FVIII* | Liver ECs | Haemophilia A | [111] |
| LV.VEC | miRT-shielded FVIII | *FVIII* | LSEC | Haemophilia A | [112] |
| **NON-VIRAL VECTORS** | | | | | |
| CaP nanoparticle | Calcium-phosphate core | *dCas9-Sirt1 RNP* | Nasal epithelium (MCT1) | Ischaemic stroke | [113] |
| PEI copolymer NP | P(MMD-co-LA)-g-PEI | *eGFP; ZNF580 plasmid* | Generic ECs | — | [114] |
| mPEG-PLGA-PEI NP | CREDVW homing peptide | *eGFP; ZNF580 plasmid* | ECs | — | [115] |
| PEG-PLGA NP | Brain-EC-target polymer | *Cas9 + gRNA (CDH5 promoter)* | Vascular ECs | Cerebrovascular disease | [116] |
| G2CNN-AAV9‡ | G2 dendrimer + EC peptides | *Cre* | Microvascular ECs | Atherosclerosis models | [117] |
| CDG2-cRGD plasmid[^120] | α_vβ_3-binding cRGD | *VEGF* | HUVECs (α_vβ_3) | Post-injury intimal thickening | [118] |

**Note:** ‡ An AAV capsid coated with a synthetic dendrimer. **Abbreviations:** ABE – adenine base editor; AdV – adenovirus; AMD – age-related macular degeneration; AVM – arteriovenous malformation; BOEC – blood-outgrowth endothelial cell; CaP – calcium phosphate; EC – endothelial cell; HSPG – heparan-sulfate proteoglycan; HUVEC – human umbilical-vein EC; LSEC – liver sinusoidal EC; LV – lentiviral vector; NP – nanoparticle; PE – prime editor; RNP – ribonucleoprotein.

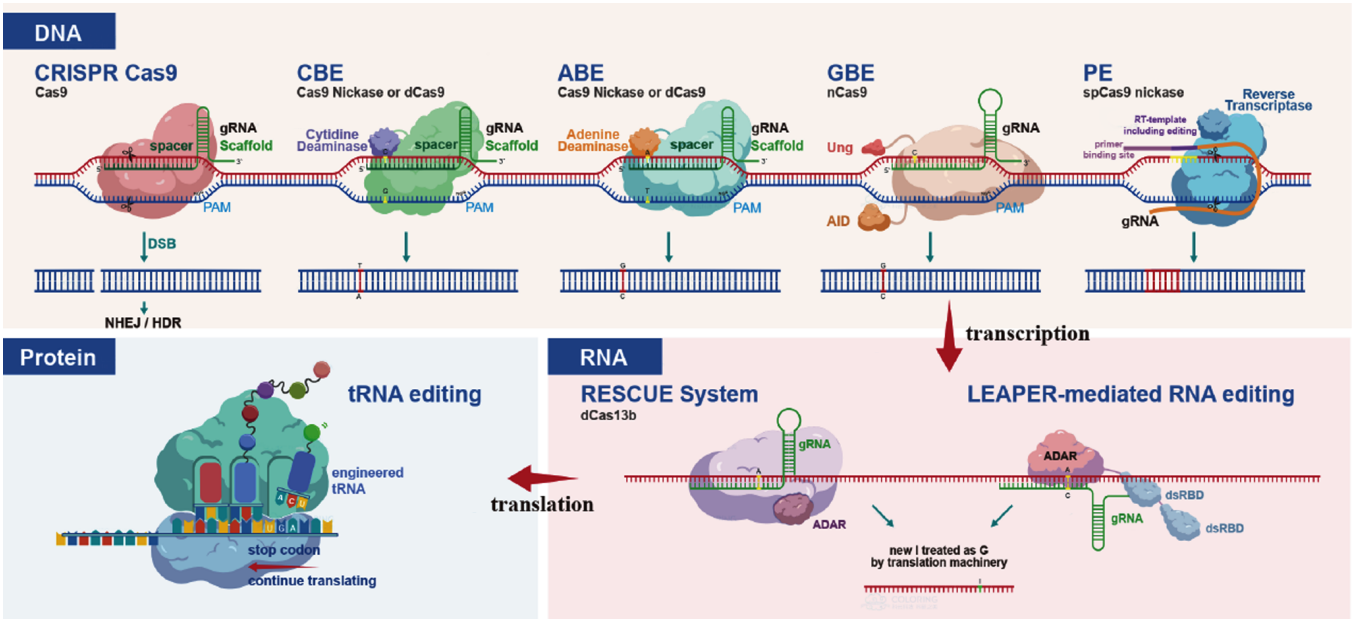


**Fig. (3).** The Working Principles of Different Gene Editing Technologies. FCCMs are associated with the functional deficiency of CCM proteins. Current approaches to address these deficiencies include DNA editing, RNA editing, and protein editing. DNA editing technology is the most advanced, with the potential to cure diseases. Although RNA and protein editing cannot yet achieve curative outcomes, they are rapidly developing and offer higher safety profiles. (*A higher resolution / colour version of this figure is available in the electronic copy of the article*).

**Table 2. Comparative analysis of gene-editing platforms for FCCM.**

| Platform | Mutation Types Addressed | Precision | Delivery Feasibility | Permanence | Suitability for Mosaic Lesions | Best use Case in FCCM |
|---|---|---|---|---|---|---|
| **CRISPR-Cas9 (DSB)** | All (including large indels or insertions) | Moderate (DSBs → NHEJ or HDR) | Challenging due to size and off-target risk | Permanent | Risk of genotoxicity; not ideal in CNS | Large insertions or knock-ins; limited CNS utility due to genotoxic risk. |
| **BE** | Transition mutations (C→T, A→G) | High (no DSBs, narrow window) | High, especially with AAV or LNP | Permanent | Excellent: partial correction can stabilize lesions | Long-term lesion stabilization; effective when ≥15–20% of mutant endothelial cells are corrected. |
| **PE** | All point mutations, small indels | Very high (programmable pegRNA) | Moderate–high, delivery is still a challenge | Permanent | Excellent: can target diverse mutations | Precise repair of complex or rare variants; ideal when multiple or compound mutations exist. |
| **Cas13 / ADAR Fusion** | A→I (RNA level only) | High (transcript-level) | High; packaging improving with smaller Cas13s | Transient | Useful for dynamic or acute modulation | Rapid, reversible transcript correction; ideal for high-risk lesions or early intervention in pediatric cases. |
| **Endogenous RNA Editors** | A→I (*via* native ADARs, no protein delivery) | Moderate (depends on ADAR levels) | Very high (no foreign protein) | Transient | Lower efficiency but low immunogenicity | Allele-specific silencing with minimal immune risk; suited for long-term modulation with repeated low-dose delivery. |

**Abbreviations:** BE – base editor; PE – prime editor; DSB – double-strand break; ADAR – adenosine deaminase acting on RNA; CNS – central nervous system; FCCM – familial cerebral cavernous malformation. "Genome altered?" refers to whether permanent DNA changes occur. "Best use-case" describes the clinical niche where each tool is most effective, particularly considering mosaic correction thresholds.

### 6.3. Glycosylase-Based Editor

To overcome the limitations of CBEs and ABEs, Glycosylase Base Editors (GBEs) have been developed to introduce non-transition substitutions, such as C-to-G, enabling the correction of a broader range of pathogenic variants [65]. GBEs use a fusion of a cytidine deaminase, a nickase Cas9, and Uracil DNA Glycosylase (UNG) to catalyze multistep base conversion through targeted DNA repair pathways [66].

Unlike standard BEs, GBEs: (i) enable purine↔pyrimidine switching, expanding the types of editable mutations; (ii) offer a narrow, high-fidelity editing window, reducing bystander effects; and (iii) maintain compatibility with dCas9 or Cas9n backbones, aiding delivery and modularity.

Experimental data have shown editing efficiencies up to 72.1% with specificity exceeding 90% in mammalian systems [67]. Integration with transcriptional Pioneer Factors has further expanded editing windows and improved access to otherwise inaccessible genomic loci [68].

For FCCM, GBEs are particularly valuable when a non-transition point mutation is identified. As with BE, correcting even a portion of the mutant endothelial cells may be enough to normalize local vascular signaling, reduce KLF2/4 activation, and prevent lesion growth.

While GBE platforms are still under active development and remain less validated than BEs, their ability to fill gaps in editing scope makes them important candidates in personalized FCCM therapy, especially when used alongside delivery methods tailored to vascular tissues [69-71].

### 6.4. Prime Editing

Prime Editing (PE) represents a versatile and highly programmable gene editing strategy capable of introducing targeted base substitutions, small insertions, and deletions without the need for Double-Strand Breaks (DSBs) or donor DNA templates. It integrates a Cas9 nickase (Cas9n) with an engineered Reverse Transcriptase (RT) and a prime editing guide RNA (pegRNA), which encodes the desired edit and directs it to the target locus [66, 72, 73]. This broad versatility is particularly relevant for FCCM, given the range of known and emerging mutations that might require more than simple transitions.

Unlike base editing, which is limited to transition mutations, PE can theoretically generate all 12 possible base-to-base conversions, as well as targeted indels. This broad editing range makes it especially promising for FCCM, which involves diverse mutations across *KRIT1*, *CCM2*, and *PDCD10*. The system's flexibility is further enhanced by its ability to introduce edits up to 34 base pairs downstream of the nicking site, increasing its reach across patient-specific mutations [74].

While PE is slightly less efficient than BE in some settings, it carries several advantages: a reduced risk of bystander edits, less susceptibility to sequence context, and a much higher edit-to-indel ratio compared to HDR-based systems [75]. In murine blastocysts, editing frequencies of 1.1%-18.5% have been reported, with precision far exceeding that of Cas9-HDR approaches [76]. While BE offers higher editing efficiencies and lower indel levels, PE provides greater versatility, precision, and targeting scope [77, 78].

In the context of FCCM, prime editing could be particularly useful in lesions with compound or multi-site mutations, or in patients whose mutations lie outside the editable windows of conventional BEs. Moreover, its potential to correct small indels or reintroduce correct splice sites aligns well with the mutation profiles observed in truncating variants.

As with other DNA-targeting methods, PE offers permanent correction, which is desirable for long-term management of FCCM, especially in pediatric cases. However, its effectiveness may vary in low-proliferation cells like brain endothelial cells. In such contexts, the capacity of PE to partially restore function in only a subset of mosaic cells may still suffice to prevent lesion progression, reduce inflammatory signaling, or restore vascular integrity.

Nonetheless, the complexity of PE's machinery—requiring multiple hybridization steps and a relatively large delivery payload—makes efficient and brain-specific delivery a major translational hurdle. Strategies involving dual-AAV systems, self-replicating RNA, or LNPs will need optimization for CNS use.

### 6.5. RNA Editing: Cas13a

Unlike DNA editing tools, RNA editors offer transient correction by targeting mutant transcripts without altering the genome. This feature is particularly attractive in the CNS, where avoiding permanent off-target damage is a key concern. In FCCM, RNA editing could serve as a safe and reversible intervention, particularly when treating lesions in critical brain regions or when long-term expression of therapeutic proteins is not yet justified.

The Cas13a and Cas13b enzymes recognize specific single-stranded RNAs *via* CRISPR RNA (crRNA) guidance, and can be fused to adenosine deaminases (*e.g.*, ADAR2) to enable A-to-I (read as G) conversions [79-81]. Fusion constructs, such as REPAIR and RESCUE, have demonstrated efficient RNA editing in mammalian cells, although early systems were large and challenging to package into AAVs.

Recent innovations have focused on minimizing Cas13 constructs for delivery efficiency. Systems like mini-Cas13X-ADAR2dd and Cas13bt are small enough to fit into single AAVs while maintaining strong editing activity and low off-target effects [82-84]. These tools have demonstrated A-to-I and C-to-U editing with high fidelity, opening doors for transient, CNS-targeted interventions with minimal risk of permanent genome disruption.

A critical challenge remains the control of collateral cleavage, a known issue with Cas13 enzymes where unintended RNAs are degraded once the enzyme is activated. High-fidelity variants, such as hfCas13d, which have been recently used to treat retinitis pigmentosa, dramatically reduce this effect [85].

In FCCM, Cas13-based RNA editing could be applied in scenarios where short-term suppression of pathogenic transcript expression may delay or reduce lesion development.

For example, transcript-level correction of mutated *KRIT1* in high-risk brainstem lesions might provide functional rescue without long-term risk. However, this approach would require repeat dosing, and delivery strategies must be tuned to reach diffuse cerebrovascular targets with high precision [85].

### 6.6. CRISPR-Free Gene Editing Tools

#### *6.6.1. Endogenous RNA Editing*

CRISPR-free RNA editing platforms offer a strategy for transient, non-permanent correction of pathogenic transcripts without modifying the genome. These tools recruit native human enzymes, such as ADAR (adenosine deaminase acting on RNA) or APOBEC, to catalyze base conversions in target mRNAs. Because no exogenous protein is introduced, immunogenicity is minimized—a major advantage in CNS applications like FCCM.

One prominent system is LEAPER (Leveraging Endogenous ADAR for Programmable Editing of RNA), first introduced in 2019. LEAPER uses short engineered RNA guides (arRNAs) to recruit endogenous ADARs to convert Adenosine (A) to Inosine (I), which is interpreted as Guanosine (G) during translation. However, early LEAPER constructs exhibited low editing efficiency and occasional bystander effects, especially when arRNA lengths were suboptimal.

To address these limitations, LEAPER 2.0 was developed, utilizing circular arRNAs (circ-arRNA) to resist exonuclease degradation and extend editing duration. Delivered *via* AAV, these guides enabled stable RNA editing in human primary cells and organoids [86, 87].

Several complementary innovations have emerged: (i) CLUSTER, a gRNA optimization platform developed by Thorsten's group, achieved up to 45% on-target editing in cell culture and 10% in mouse liver with minimal off-target activity; (ii) cadRNA, developed by Prashant Mali's lab, uses highly stable circular guide RNAs and has demonstrated 53% editing of the *PCSK9* transcript and 12% correction of a nonsense mutation in *IDUA* in mouse models of mucopolysaccharidosis; (iii) In primates, AIMers delivered *via* GalNAc conjugation enabled 50% editing of liver transcripts without bystander effects and persisted for over a month [88-90].

Despite these promising results, translation to FCCM remains challenging. The CNS vasculature presents a more complex environment, where achieving efficient, lesion-specific delivery and sufficient editing coverage across mosaic lesions is more difficult. Furthermore, RNA editing is inherently transient, requiring repeated administration for sustained therapeutic benefit.

Still, endogenous editing holds key advantages: no vector-encoded proteins, lower immune activation, and reversible modulation of gene function. For FCCM, particularly in high-risk or surgically inoperable lesions, transient transcript correction could reduce lesion activity or hemorrhagic risk without risking permanent off-target damage. Future improvements in cell-type-selective delivery and editing efficiency in the CNS endothelium will determine clinical viability [91].

#### *6.6.2. Editing Tools for Nonsense Mutations*

Although missense and truncating mutations are more prevalent in FCCM, nonsense mutations that introduce Premature Termination Codons (PTCs) also contribute to pathogenesis in some patients. These can produce non-functional, truncated proteins, making them attractive targets for correction *via* stop codon readthrough technologies.

One such tool is RESTART, a programmable RNA base editor developed by Yi Chengqi's team. RESTART introduces pseudouridine modifications at PTCs, enabling ribosomes to bypass stop signals and restore translation of full-length proteins [92]. This strategy has been validated in mammalian cells and could be applicable to FCCM variants that truncate *KRIT1*, *CCM2*, or *PDCD10*.

Another approach involves engineered tRNA/tRNA-synthetase pairs (tRNA-aaRS) that suppress stop codons by inserting the correct amino acid during translation. In 2022, Xia Qing's team demonstrated that AAV-delivered tRNA-aaRS complexes restored dystrophin expression in both murine and patient-derived muscle cells, leading to functional rescue in Duchenne muscular dystrophy models [93].

These strategies do not edit the underlying DNA or RNA, making them non-mutagenic and theoretically repeatable with minimal genomic risk. However, key challenges remain: (i) Specificity: Ensuring readthrough only occurs at pathogenic PTCs without misreading normal stop codons; (ii) Efficiency: Achieving sufficient protein restoration to reverse pathology; (iii) Delivery: Targeting brain endothelial cells, which are dispersed and often embedded in lesion cores.

Given that PTC mutations are relatively rare in FCCM compared to missense variants, the immediate application of these tools is limited. Nonetheless, they provide an additional therapeutic avenue and may complement DNA/RNA editing in future multi-modal strategies. For FCCM cases harboring nonsense mutations, tRNA-based or pseudouridine-mediated readthrough could enable partial protein rescue, potentially mitigating lesion progression without needing full genetic correction.

## 7. OUTLOOK AND CHALLENGES

The localized nature of Cerebral Cavernous Malformations (CCMs) may offer some convenience for drug administration. However, the mosaic structure, multifocal distribution, and delicate location of CCMs in the Central Nervous System (CNS) also create formidable challenges for safety and therapeutic delivery. Although CCM lesions are focal, not all cells harbor the disease-causing mutations. This mosaicism complicates treatment strategies, as partial correction of mutant cells may be insufficient to curb disease progression, and uniform editing across all affected endothelial cells remains difficult. In many disease models, surpassing a minimum threshold of corrected cells can stabilize the pathology; however, for Familial Cerebral Cavernous Malformations (FCCMs), determining this threshold requires advanced *in vivo* models capable of mirroring the human lesion's heterogeneity.

Equally pressing is the inflammatory dimension of CCM pathology. Loss-of-function mutations in *KRIT1*, CCM2, or *PDCD10* can compromise the endothelium's antioxidant and anti-inflammatory defenses, allowing neuroinflammatory astrocytes and other immune mediators to exacerbate lesion growth. Links between inflamed CCM endothelium, leukocyte recruitment, and immunothrombosis indicate that neuroinflammation plays a central role in lesion evolution. Thus, a dual therapeutic approach—integrating precise gene editing with anti-inflammatory interventions—may reduce lesion burden and hemorrhagic risk more effectively than gene manipulation alone.

From a technical perspective, FCCM presents a compelling testbed for gene editing, given that it stems from relatively well-defined, monogenic disruptions. Multiple platforms (DNA-targeting edits, RNA-based modifications, and transcriptional modulation) are under active development. Nonetheless, each option must achieve high editing precision within fragile endothelial cells, where off-target events could spur severe neurological consequences. Preclinical models—especially those capable of single-base or prime editing—show promise but must be refined to address delivery efficiency, minimize off-target effects, and ensure the durability of gene correction.

The delivery question remains especially critical. Crossing the Blood-Brain Barrier (BBB) without damaging it requires careful selection or engineering of viral and non-viral vectors. Lipid nanoparticles, exosomes, and novel AAV serotypes with enhanced BBB permeability may be harnessed. Concurrently, strategies for transient BBB disruption might be explored, albeit with caution to avoid inadvertently promoting hemorrhage. Achieving widespread transduction across multiple vascular sites, ensuring adequate dose accumulation, and addressing potential immune neutralization during re-administration all demand continued innovation.

In parallel, personalized therapy becomes increasingly relevant as next-generation sequencing clarifies the genetic heterogeneity of FCCM patients. Accurate diagnostic tools that identify each patient's mutation profile can guide the choice of gene editor (*e.g.*, single-base editors *vs*. prime editors), further elevating treatment efficacy and mitigating off-target concerns. Additionally, combination therapies that stabilize endothelial junctions or dampen inflammatory cascades may synergize with gene correction, offering more comprehensive outcomes and improving patient quality of life.

Despite these technical hurdles, gene editing remains a transformative prospect for FCCM. Advanced editing systems have already generated single-base knock-in models that facilitate in-depth pathophysiological studies, enabling rigorous testing of both efficacy and safety. However, scaling up from bench to bedside necessitates overcoming major regulatory and manufacturing hurdles, ensuring consistent vector production, meeting stringent safety standards, and navigating ethical considerations tied to editing the CNS. Immune responses to repeated dosing, as well as the potential for late-arising genetic alterations, also mandate strict long-term monitoring.

Overall, continued progress in gene editing tools and delivery systems, alongside a deeper understanding of CCM biology, will be pivotal for effective FCCM therapies. By successfully targeting the core genetic deficits underlying CCM lesions, clinicians could one day offer a non-surgical, neuron-sparing alternative that not only lowers hemorrhagic risk but also improves long-term neurological outcomes. Should these strategies succeed, FCCM may stand as a pioneering example for treating other monogenic or multifactorial cerebrovascular diseases, ultimately ushering in a new era of personalized, precision medicine for complex CNS disorders.

## CONCLUSION

This review highlights the promising potential of gene therapy for FCCM, a monogenic disorder driven by mutations in *KRIT1*, *CCM2,* or *PDCD10*. Existing evidence shows that gene-editing tools, such as BE and PE, can precisely correct pathogenic mutations without inducing double-strand breaks, offering a non-surgical strategy to restore vascular endothelial function. Preclinical studies indicate that partial correction of mutant endothelial cells (as low as 15–30%) in mosaic lesions may be sufficient to stabilize vascular barriers and suppress lesion progression, underscoring the feasibility of targeted gene repair.

Delivery systems, particularly AAVs and LNPs, have advanced to enable CNS targeting, though challenges remain in crossing the BBB and achieving uniform transduction across multifocal lesions. The mosaic nature of FCCM lesions and potential immune responses to viral vectors further necessitate optimization of delivery strategies. Additionally, the inflammatory cascade associated with CCM pathogenesis highlights the need for combinatorial approaches that integrate gene editing with anti-inflammatory therapies.

While gene therapy holds transformative promise for FCCM, critical hurdles must be addressed, including defining the minimal threshold of corrected cells required for clinical efficacy, minimizing off-target effects in sensitive CNS tissues, and developing durable, safe delivery methods. Translational success will depend on integrating precision editing with personalized medicine, leveraging patient-specific mutation profiles to tailor therapeutic approaches. Ultimately, overcoming these challenges may establish gene therapy as a cornerstone for treating FCCM and other monogenic cerebrovascular disorders.

## AUTHORS' CONTRIBUTIONS

The authors confirm their contribution to the paper as follows: Writing – original draft: KM and MR; Visualization: KM and MR; Writing – review & editing: KM, MR, SL, YW, CY; Literature search & data curation: YX; Formal analysis: YW, WL, YD; Validation: YW, WL, YD; Supervision: GL and JY; Conceptualization: CY; Funding acquisition: CY. KM and MR contributed equally to this work. CY is the corresponding author. All authors read and approved the final manuscript.

## LIST OF ABBREVIATIONS

| | | |
|---|---|---|
| ABE | = | Adenine Base Editor |
| AA | = | Amino Acid |
| ADAR | = | Adenosine Deaminase Acting on RNA |
| AAV | = | Adeno-Associated Virus |
| AIMers | = | Antisense-Interfering Molecules |
| ANK | = | Ankyrin |
| APOBEC | = | Apolipoprotein B mRNA Editing Catalytic Polypeptide |
| AVM | = | Arteriovenous Malformation |
| BBB | = | Blood-Brain Barrier |
| BE | = | Base Editing |
| CaP | = | Calcium Phosphate |
| Cas9n | = | Cas9 Nickase |
| CBE | = | Cytidine Base Editor |
| CNS | = | Central Nervous System |
| crRNA | = | CRISPR RNA |
| DSB | = | Double-Strand Break |
| EMT | = | Endothelial-to-Mesenchymal Transition |
| FCCM | = | Familial Cerebral Cavernous Malformation |
| FERM | = | Four-point-one, Ezrin, Radixin, Moesin Domain |
| HDR | = | Homology-Directed Repair |
| HF | = | High Fidelity |
| HSPG | = | Heparan Sulfate Proteoglycan |
| IDUA | = | Iduronidase Alpha-L |
| ICAP1α | = | Integrin Cytoplasmic Domain-Associated Protein-1 Alpha |
| Indel | = | Insertion or Deletion |
| KRIT1 | = | Krev-1 Interaction Trapped-1 |
| LEAPER | = | Leveraging Endogenous ADAR for Programmable Editing of RNA |
| LNP | = | Lipid Nanoparticle |
| LV | = | Lentiviral Vector |
| MAP3K3 | = | Mitogen-Activated Protein Kinase Kinase Kinase 3 |
| MRI | = | Magnetic Resonance Imaging |
| mRNA | = | Messenger RNA |
| NBE | = | Next-Generation Base Editor |
| NHEJ | = | Non-Homologous End Joining |
| NP | = | Nanoparticle |
| NPXY | = | Asn-Pro-X-Tyr Motif |
| pamRNA | = | Prime Editing gRNA Scaffold |
| PBS | = | Primer Binding Site |
| PCSK9 | = | Proprotein Convertase Subtilisin/Kexin Type 9 |
| PDCD10 | = | Programmed Cell Death 10 |
| PE | = | Prime Editing |
| pegRNA | = | Prime Editing Guide RNA |
| PTC | = | Premature Termination Codon |
| RAP1A | = | Ras-Associated Protein 1A |
| REPAIR | = | RNA Editing for Programmable A-to-I Replacement |
| RESCUE | = | RNA Editing for Specific C-to-U Exchange |
| RHOA | = | Ras Homolog Family Member A |
| RNP | = | Ribonucleoprotein |
| ROCK | = | Rho-Associated Protein Kinase |
| RT | = | Reverse Transcriptase |
| sgRNA | = | Single Guide RNA |
| siRNA | = | Small Interfering RNA |
| tRNA | = | Transfer RNA |
| UNG | = | Uracil DNA Glycosylase |
| ZFNs | = | Zinc Finger Nucleases |

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## CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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## REFERENCES


[1] Flemming KD, Graff-Radford J, Aakre J, *et al.* Population-based prevalence of cerebral cavernous malformations in older adults. JAMA Neurol 2017; 74(7): 801-5. http://dx.doi.org/10.1001/jamaneurol.2017.0439 PMID: 28492932

[2] Dalyai RT, Ghobrial G, Awad I, *et al.* Management of incidental cavernous malformations: A review. Neurosurg Focus 2011; 31(6): E5. http://dx.doi.org/10.3171/2011.9.FOCUS11211 PMID: 22133177

[3] Lanfranconi S, Scola E, Meessen JMTA, *et al.* Safety and efficacy of propranolol for treatment of familial cerebral cavernous malformations (Treat_CCM): A randomised, open-label, blinded-endpoint, phase 2 pilot trial. Lancet Neurol 2023; 22(1): 35-44. http://dx.doi.org/10.1016/S1474-4422(22)00409-4 PMID: 36403580

[4] Santos AN, Rauschenbach L, Saban D, *et al.* Medication intake and hemorrhage risk in patients with familial cerebral cavernous malformations. J Neurosurg 2022; 137(4): 1088-94. http://dx.doi.org/10.3171/2022.1.JNS212724 PMID: 35213840

[5] Kim S, Moon J, Jung KH, *et al.* Clinicoradiologic data of familial cerebral cavernous malformation with age-related disease burden. Ann Clin Transl Neurol 2023; 10(3): 373-83. http://dx.doi.org/10.1002/acn3.51728 PMID: 36629374

[6] Li X, Fisher OS, Boggon TJ. The cerebral cavernous malformations proteins. Oncotarget 2015; 6(32): 32279-80. http://dx.doi.org/10.18632/oncotarget.5443 PMID: 26356566

[7] Choquet H, Trapani E, Goitre L, *et al.* Cytochrome P450 and matrix metalloproteinase genetic modifiers of disease severity in Cerebral Cavernous Malformation type 1. Free Radic Biol Med 2016; 92: 100-9. http://dx.doi.org/10.1016/j.freeradbiomed.2016.01.008 PMID: 26795600

[8] Tang AT, Sullivan KR, Hong CC, *et al.* Distinct cellular roles for PDCD10 define a gut-brain axis in cerebral cavernous malformation. Sci Transl Med 2019; 11(520): eaaw3521. http://dx.doi.org/10.1126/scitranslmed.aaw3521 PMID: 31776290

[9] Yang C, Nicholas VHL, Zhao J, *et al.* A novel CCM1/KRIT1 heterozygous nonsense mutation (c.1864C>T) associated with familial cerebral cavernous malformation: A genetic insight from an 8-year continuous observational study. J Mol Neurosci 2017; 61(4): 511-23. http://dx.doi.org/10.1007/s12031-017-0893-1 PMID: 28255959

[10] Yang C, Zhao J, Wu B, Zhong H, Li Y, Xu Y. Identification of a novel deletion mutation (c.1780delG) and a novel splice-site mutation (c.1412-1G>A) in the CCM1/KRIT1 gene associated with familial cerebral cavernous malformation in the chinese population. J Mol Neurosci 2017; 61(1): 8-15. http://dx.doi.org/10.1007/s12031-016-0836-2 PMID: 27649701

[11] Yang C, Wu B, Zhong H, Li Y, Zheng X, Xu Y. A novel CCM1/KRIT1 heterozygous deletion mutation (c.1919delT) in a Chinese family with familial cerebral cavernous malformation. Clin Neurol Neurosurg 2018; 164: 44-6. http://dx.doi.org/10.1016/j.clineuro.2017.11.005 PMID: 29169046

[12] Wang X, Liu X, Lee N, *et al.* Features of a Chinese family with cerebral cavernous malformation induced by a novel CCM1 gene mutation. Chin Med J 2013; 126(18): 3427-32. http://dx.doi.org/10.3760/cma.j.issn.0366-6999.20130590 PMID: 24034083

[13] Liu W, Liu M, Lu D, *et al.* A Chinese family with cerebral cavernous malformation caused by a frameshift mutation of the *CCM1* gene: A Case report and review of the literature. Front Neurol 2022; 13: 795514. http://dx.doi.org/10.3389/fneur.2022.795514 PMID: 35444609

[14] Wang H, Pan Y, Zhang Z, *et al.* A novel KRIT1/CCM1 gene insertion mutation associated with cerebral cavernous malformations in a Chinese family. J Mol Neurosci 2017; 61(2): 221-6. http://dx.doi.org/10.1007/s12031-017-0881-5 PMID: 28160210

[15] Zhao Y, Xie L, Li P, *et al.* A novel CCM1 gene mutation causes cerebral cavernous malformation in a Chinese family. J Clin Neurosci 2011; 18(1): 61-5. http://dx.doi.org/10.1016/j.jocn.2010.04.051 PMID: 20884211

[16] Bergametti F, Denier C, Labauge P, *et al.* Mutations within the programmed cell death 10 gene cause cerebral cavernous malformations. Am J Hum Genet 2005; 76(1): 42-51. http://dx.doi.org/10.1086/426952 PMID: 15543491

[17] Couteulx SL, Jung HH, Labauge P, *et al.* Truncating mutations in CCM1, encoding KRIT1, cause hereditary cavernous angiomas. Nat Genet 1999; 23(2): 189-93. http://dx.doi.org/10.1038/13815 PMID: 10508515

[18] Fisher OS, Boggon TJ. Signaling pathways and the cerebral cavernous malformations proteins: Lessons from structural biology. Cell Mol Life Sci 2014; 71(10): 1881-92. http://dx.doi.org/10.1007/s00018-013-1532-9 PMID: 24287896

[19] Li X, Zhang R, Zhang H, *et al.* Crystal structure of CCM3, a cerebral cavernous malformation protein critical for vascular integrity. J Biol Chem 2010; 285(31): 24099-107. http://dx.doi.org/10.1074/jbc.M110.128470 PMID: 20489202

[20] Zhou Z, Tang AT, Wong WY, *et al.* Erratum: Corrigendum: Cerebral cavernous malformations arise from endothelial gain of MEKK3–KLF2/4 signalling. Nature 2016; 536(7617): 488. http://dx.doi.org/10.1038/nature18311 PMID: 27281211

[21] Weinsheimer S, Nelson J, Abla AA, *et al.* Intracranial hemorrhage rate and lesion burden in patients with familial cerebral cavernous malformation. J Am Heart Assoc 2023; 12(3): 027572. http://dx.doi.org/10.1161/JAHA.122.027572 PMID: 36695309

[22] Akers A, Al-Shahi Salman R, A Awad I, *et al.* Synopsis of guidelines for the clinical management of cerebral cavernous malformations: Consensus recommendations based on systematic literature review by the angioma alliance scientific advisory board clinical experts panel. Neurosurgery 2017; 80(5): 665-80. http://dx.doi.org/10.1093/neuros/nyx091 PMID: 28387823

[23] Petersen TA, Morrison LA, Schrader RM, Hart BL. Familial *versus* sporadic cavernous malformations: Differences in developmental venous anomaly association and lesion phenotype. AJNR Am J Neuroradiol 2010; 31(2): 377-82. http://dx.doi.org/10.3174/ajnr.A1822 PMID: 19833796

[24] Moore SA, Brown RD, Christianson TJH, Flemming KD. Long-term natural history of incidentally discovered cavernous malformations in a single-center cohort. J Neurosurg 2014; 120(5): 1188-92. http://dx.doi.org/10.3171/2014.1.JNS131619 PMID: 24628608

[25] Fox CK, Nelson J, McCulloch CE, *et al.* Seizure incidence rates in Children and adults with familial cerebral cavernous malformations. Neurology 2021; 97(12): e1210-6. http://dx.doi.org/10.1212/WNL.0000000000012569 PMID: 34389651

[26] Haasdijk RA, Cheng C, Maat-Kievit AJ, Duckers HJ. Cerebral cavernous malformations: From molecular pathogenesis to genetic counselling and clinical management. Eur J Hum Genet 2012; 20(2): 134-40. http://dx.doi.org/10.1038/ejhg.2011.155 PMID: 21829231

[27] Mouchtouris N, Chalouhi N, Chitale A, *et al.* Management of cerebral cavernous malformations: From diagnosis to treatment. ScientificWorldJournal 2015; 2015(1): 808314. http://dx.doi.org/10.1155/2015/808314 PMID: 25629087

[28] Al-Shahi R, Bhattacharya JJ, Currie DG, *et al.* Prospective, population-based detection of intracranial vascular malformations in adults: The Scottish intracranial vascular malformation study (SIVMS). Stroke 2003; 34(5): 1163-9. http://dx.doi.org/10.1161/01.STR.0000069018.90456.C9 PMID: 12702837

[29] Richardson BT, Dibble CF, Borikova AL, Johnson GL. Cerebral cavernous malformation is a vascular disease associated with activated RhoA signaling. bchm 2013; 394(1): 35-42. http://dx.doi.org/10.1515/hsz-2012-0243 PMID: 23096573

[30] Polster SP, Stadnik A, Akers AL, *et al.* Atorvastatin treatment of cavernous angiomas with symptomatic hemorrhage exploratory proof of concept (AT CASH EPOC) trial. Neurosurgery 2019; 85(6): 843-53. http://dx.doi.org/10.1093/neuros/nyy539 PMID: 30476251

[31] Li W, Shenkar R, Detter MR, *et al.* Propranolol inhibits cavernous vascular malformations by β1 adrenergic receptor antagonism in animal models. J Clin Invest 2021; 131(3): 144893. http://dx.doi.org/10.1172/JCI144893 PMID: 33301422

[32] Shenkar R, Moore T, Benavides C, *et al.* Propranolol as therapy for cerebral cavernous malformations: A cautionary note. J Transl Med 2022; 20(1): 160. http://dx.doi.org/10.1186/s12967-022-03360-4 PMID: 35382850

[33] Sahoo T, Johnson EW, Thomas JW, *et al.* Mutations in the gene encoding KRIT1, a Krev-1/rap1a binding protein, cause cerebral cavernous malformations (CCM1). Hum Mol Genet 1999; 8(12): 2325-33.
http://dx.doi.org/10.1093/hmg/8.12.2325 PMID: 10545614

[34] Zhang J, Basu S, Rigamonti D, Dietz HC, Clatterbuck RE. KRIT1 modulates β1-integrin-mediated endothelial cell proliferatION. Neurosurgery 2008; 63(3): 571-8.
http://dx.doi.org/10.1227/01.NEU.0000325255.30268.B0 PMID: 18812969

[35] Swamy H, Glading AJ. Contribution of protein–protein interactions to the endothelial-barrier-stabilizing function of KRIT1. J Cell Sci 2022; 135(2): jcs258816.
http://dx.doi.org/10.1242/jcs.258816 PMID: 34918736

[36] Zhang J, Clatterbuck RE, Rigamonti D, Chang DD, Dietz HC. Interaction between krit1 and icap1alpha infers perturbation of integrin beta1-mediated angiogenesis in the pathogenesis of cerebral cavernous malformation. Hum Mol Genet 2001; 10(25): 2953-60.
http://dx.doi.org/10.1093/hmg/10.25.2953 PMID: 11741838

[37] Tebas P, Stein D, Tang WW, *et al.* Gene editing of CCR5 in autologous CD4 T cells of persons infected with HIV. N Engl J Med 2014; 370(10): 901-10.
http://dx.doi.org/10.1056/NEJMoa1300662 PMID: 24597865

[38] Shahryari A, Saghaeian Jazi M, Mohammadi S, *et al.* Development and clinical translation of approved gene therapy products for genetic disorders. Front Genet 2019; 10: 868.
http://dx.doi.org/10.3389/fgene.2019.00868 PMID: 31608113

[39] Ma CC, Wang ZL, Xu T, He ZY, Wei YQ. The approved gene therapy drugs worldwide: From 1998 to 2019. Biotechnol Adv 2020; 40: 107502.
http://dx.doi.org/10.1016/j.biotechadv.2019.107502 PMID: 31887345

[40] Li T, Yang Y, Qi H, *et al.* CRISPR/Cas9 therapeutics: Progress and prospects. Signal Transduct Target Ther 2023; 8(1): 36.
http://dx.doi.org/10.1038/s41392-023-01309-7 PMID: 36646687

[41] Zafar A, Quadri SA, Farooqui M, *et al.* Familial cerebral cavernous malformations. Stroke 2019; 50(5): 1294-301.
http://dx.doi.org/10.1161/STROKEAHA.118.022314 PMID: 30909834

[42] Doudna JA, Charpentier E. The new frontier of genome engineering with CRISPR-Cas9. Science 2014; 346(6213): 1258096.
http://dx.doi.org/10.1126/science.1258096 PMID: 25430774

[43] Mendell JR, Al-Zaidy SA, Rodino-Klapac LR, *et al.* Current clinical applications of *in vivo* gene Therapy with AAVs. Mol Ther 2021; 29(2): 464-88.
http://dx.doi.org/10.1016/j.ymthe.2020.12.007 PMID: 33309881

[44] Breda L, Papp TE, Triebwasser MP, *et al. In vivo* hematopoietic stem cell modification by mRNA delivery. Science 2023; 381(6656): 436-43.
http://dx.doi.org/10.1126/science.ade6967 PMID: 37499029

[45] Ferrari S, Naldini L. A step toward stem cell engineering *in vivo*. Science 2023; 381(6656): 378-9.
http://dx.doi.org/10.1126/science.adj0997 PMID: 37499013

[46] Xu Y, Shrestha N, Préat V, Beloqui A. An overview of *in vitro*, *ex vivo* and *in vivo* models for studying the transport of drugs across intestinal barriers. Adv Drug Deliv Rev 2021; 175: 113795.
http://dx.doi.org/10.1016/j.addr.2021.05.005 PMID: 33989702

[47] Krolak T, Chan KY, Kaplan L, *et al.* A high-efficiency AAV for endothelial cell transduction throughout the central nervous system. Nat Cardiovasc Res 2022; 1(4): 389-400.
http://dx.doi.org/10.1038/s44161-022-00046-4 PMID: 35571675

[48] Chen F, Alphonse M, Liu Q. Strategies for nonviral nanoparticle-based delivery of CRISPR/Cas9 therapeutics. Wiley Interdiscip Rev Nanomed Nanobiotechnol 2020; 12(3): 1609.
http://dx.doi.org/10.1002/wnan.1609 PMID: 31797562

[49] Weng J, Yang Y, Song D, *et al.* Somatic MAP3K3 mutation defines a subclass of cerebral cavernous malformation. Am J Hum Genet 2021; 108(5): 942-50.
http://dx.doi.org/10.1016/j.ajhg.2021.04.005 PMID: 33891857

[50] Yang H, Ren S, Yu S, *et al.* Methods favoring homology-directed repair choice in response to CRISPR/Cas9 induced-double strand breaks. Int J Mol Sci 2020; 21(18): 6461.
http://dx.doi.org/10.3390/ijms21186461 PMID: 32899704

[51] Shen S, Loh TJ, Shen H, Zheng X, Shen H. CRISPR as a strong gene editing tool. BMB Rep 2017; 50(1): 20-4.
http://dx.doi.org/10.5483/BMBRep.2017.50.1.128 PMID: 27616359

[52] Ran FA, Hsu PD, Lin CY, *et al.* Double nicking by RNA-guided CRISPR Cas9 for enhanced genome editing specificity. Cell 2013; 154(6): 1380-9.
http://dx.doi.org/10.1016/j.cell.2013.08.021 PMID: 23992846

[53] Wu Y, Zhou H, Fan X, *et al.* Correction of a genetic disease by CRISPR-Cas9-mediated gene editing in mouse spermatogonial stem cells. Cell Res 2015; 25(1): 67-79.
http://dx.doi.org/10.1038/cr.2014.160 PMID: 25475058

[54] Fu YW, Dai XY, Wang WT, *et al.* Dynamics and competition of CRISPR–Cas9 ribonucleoproteins and AAV donor-mediated NHEJ, MMEJ and HDR editing. Nucleic Acids Res 2021; 49(2): 969-85.
http://dx.doi.org/10.1093/nar/gkaa1251 PMID: 33398341

[55] Ran FA, Hsu PD, Wright J, Agarwala V, Scott DA, Zhang F. Genome engineering using the CRISPR-Cas9 system. Nat Protoc 2013; 8(11): 2281-308.
http://dx.doi.org/10.1038/nprot.2013.143 PMID: 24157548

[56] Zetsche B, Gootenberg JS, Abudayyeh OO, *et al.* Cpf1 is a single RNA-guided endonuclease of a class 2 CRISPR-Cas system. Cell 2015; 163(3): 759-71.
http://dx.doi.org/10.1016/j.cell.2015.09.038 PMID: 26422227

[57] Mao Z, Chen R, Wang X, *et al.* CRISPR/Cas12a-based technology: A powerful tool for biosensing in food safety. Trends Food Sci Technol 2022; 122: 211-22.
http://dx.doi.org/10.1016/j.tifs.2022.02.030 PMID: 35250172

[58] Komor AC, Kim YB, Packer MS, Zuris JA, Liu DR. Programmable editing of a target base in genomic DNA without double-stranded DNA cleavage. Nature 2016; 533(7603): 420-4.
http://dx.doi.org/10.1038/nature17946 PMID: 27096365

[59] Landrum MJ, Lee JM, Benson M, *et al.* ClinVar: Public archive of interpretations of clinically relevant variants. Nucleic Acids Res 2016; 44(D1): D862-8.
http://dx.doi.org/10.1093/nar/gkv1222 PMID: 26582918

[60] Koblan LW, Doman JL, Wilson C, *et al.* Improving cytidine and adenine base editors by expression optimization and ancestral reconstruction. Nat Biotechnol 2018; 36(9): 843-6.
http://dx.doi.org/10.1038/nbt.4172 PMID: 29813047

[61] Lee HK, Oh Y, Hong J, Lee SH, Hur JK. Development of CRISPR technology for precise single-base genome editing: A brief review. BMB Rep 2021; 54(2): 98-105.
http://dx.doi.org/10.5483/BMBRep.2021.54.2.217 PMID: 33298245

[62] Gaudelli NM, Komor AC, Rees HA, *et al.* Programmable base editing of A•T to G•C in genomic DNA without DNA cleavage. Nature 2017; 551(7681): 464-71.
http://dx.doi.org/10.1038/nature24644 PMID: 29160308

[63] Richter MF, Zhao KT, Eton E, *et al.* Phage-assisted evolution of an adenine base editor with improved Cas domain compatibility and activity. Nat Biotechnol 2020; 38(7): 883-91.
http://dx.doi.org/10.1038/s41587-020-0453-z PMID: 32433547

[64] Liang P, Huang J. Off-target challenge for base editor-mediated genome editing. Cell Biol Toxicol 2019; 35(3): 185-7.
http://dx.doi.org/10.1007/s10565-019-09474-8 PMID: 31041571

[65] Lu C, Kuang J, Shao T, *et al.* Prime editing: An all-rounder for genome editing. Int J Mol Sci 2022; 23(17): 9862.
http://dx.doi.org/10.3390/ijms23179862 PMID: 36077252

[66] Zhao D, Li J, Li S, *et al.* Glycosylase base editors enable C-to-A and C-to-G base changes. Nat Biotechnol 2021; 39(1): 35-40.
http://dx.doi.org/10.1038/s41587-020-0592-2 PMID: 32690970

[67] Sun N, Zhao D, Li S, Zhang Z, Bi C, Zhang X. Reconstructed glycosylase base editors GBE2.0 with enhanced C-to-G base editing efficiency and purity. Mol Ther 2022; 30(7): 2452-63.
http://dx.doi.org/10.1016/j.ymthe.2022.03.023 PMID: 35381364

[68] Yang C, Dong X, Ma Z, Li B, Bi C, Zhang X. Pioneer factor improves crispr-based C-To-G and C-To-T base editing. Adv Sci 2022; 9(26): 2202957.
http://dx.doi.org/10.1002/advs.202202957 PMID: 35861371

[69] Yeh WH, Shubina-Oleinik O, Levy JM, *et al. In vivo* base editing restores sensory transduction and transiently improves auditory

function in a mouse model of recessive deafness. Sci Transl Med 2020; 12(546): eaay9101.
http://dx.doi.org/10.1126/scitranslmed.aay9101 PMID: 32493795

[70] Koblan LW, Erdos MR, Wilson C, *et al. In vivo* base editing rescues Hutchinson–Gilford progeria syndrome in mice. Nature 2021; 589(7843): 608-14.
http://dx.doi.org/10.1038/s41586-020-03086-7 PMID: 33408413

[71] Ryu SM, Koo T, Kim K, *et al.* Adenine base editing in mouse embryos and an adult mouse model of Duchenne muscular dystrophy. Nat Biotechnol 2018; 36(6): 536-9.
http://dx.doi.org/10.1038/nbt.4148 PMID: 29702637

[72] Kurt IC, Zhou R, Iyer S, *et al.* CRISPR C-to-G base editors for inducing targeted DNA transversions in human cells. Nat Biotechnol 2021; 39(1): 41-6.
http://dx.doi.org/10.1038/s41587-020-0609-x PMID: 32690971

[73] Chen L, Park JE, Paa P, *et al.* Programmable C:G to G:C genome editing with CRISPR-Cas9-directed base excision repair proteins. Nat Commun 2021; 12(1): 1384.
http://dx.doi.org/10.1038/s41467-021-21559-9 PMID: 33654077

[74] Scholefield J, Harrison PT. Prime editing – An update on the field. Gene Ther 2021; 28(7-8): 396-401.
http://dx.doi.org/10.1038/s41434-021-00263-9 PMID: 34031549

[75] Anzalone AV, Randolph PB, Davis JR, *et al.* Search-and-replace genome editing without double-strand breaks or donor DNA. Nature 2019; 576(7785): 149-57.
http://dx.doi.org/10.1038/s41586-019-1711-4 PMID: 31634902

[76] Liu Y, Li X, He S, *et al.* Efficient generation of mouse models with the prime editing system. Cell Discov 2020; 6(1): 27.
http://dx.doi.org/10.1038/s41421-020-0165-z PMID: 32351707

[77] Liang R, He Z, Zhao KT, *et al.* Prime editing using CRISPR-Cas12a and circular RNAs in human cells. Nat Biotechnol 2024; 42(12): 1867-75.
http://dx.doi.org/10.1038/s41587-023-02095-x PMID: 38200119

[78] Yan J, Oyler-Castrillo P, Ravisankar P, *et al.* Improving prime editing with an endogenous small RNA-binding protein. Nature 2024; 628(8008): 639-47.
http://dx.doi.org/10.1038/s41586-024-07259-6 PMID: 38570691

[79] Abudayyeh OO, Gootenberg JS, Konermann S, *et al.* C2c2 is a single-component programmable RNA-guided RNA-targeting CRISPR effector. Science 2016; 353(6299): aaf5573.
http://dx.doi.org/10.1126/science.aaf5573 PMID: 27256883

[80] Behr M, Zhou J, Xu B, Zhang H. *In vivo* delivery of CRISPR-Cas9 therapeutics: Progress and challenges. Acta Pharm Sin B 2021; 11(8): 2150-71.
http://dx.doi.org/10.1016/j.apsb.2021.05.020 PMID: 34522582

[81] Smargon AA, Cox DBT, Pyzocha NK, *et al.* Cas13b is a type VI-B CRISPR-associated RNA-guided rnase differentially regulated by accessory proteins Csx27 and Csx28. Mol Cell 2017; 65(4): 618-630.e7.
http://dx.doi.org/10.1016/j.molcel.2016.12.023 PMID: 28065598

[82] Konermann S, Lotfy P, Brideau NJ, Oki J, Shokhirev MN, Hsu PD. Transcriptome engineering with RNA-targeting type VI-D CRISPR effectors. Cell 2018; 173(3): 665-676.e14.
http://dx.doi.org/10.1016/j.cell.2018.02.033 PMID: 29551272

[83] Kannan S, Altae-Tran H, Jin X, *et al.* Compact RNA editors with small Cas13 proteins. Nat Biotechnol 2022; 40(2): 194-7.
http://dx.doi.org/10.1038/s41587-021-01030-2 PMID: 34462587

[84] Xu C, Zhou Y, Xiao Q, *et al.* Programmable RNA editing with compact CRISPR–Cas13 systems from uncultivated microbes. Nat Methods 2021; 18(5): 499-506.
http://dx.doi.org/10.1038/s41592-021-01124-4 PMID: 33941935

[85] Yan Z, Yao Y, Li L, *et al.* Treatment of autosomal dominant retinitis pigmentosa caused by RHO-P23H mutation with high-fidelity Cas13X in mice. Mol Ther Nucleic Acids 2023; 33: 750-61.
http://dx.doi.org/10.1016/j.omtn.2023.08.002 PMID: 37621413

[86] Nishikura K. Functions and regulation of RNA editing by ADAR deaminases. Annu Rev Biochem 2010; 79(1): 321-49.
http://dx.doi.org/10.1146/annurev-biochem-060208-105251 PMID: 20192758

[87] Yi Z, Qu L, Tang H, *et al.* Engineered circular ADAR-recruiting RNAs increase the efficiency and fidelity of RNA editing *in vitro* and *in vivo*. Nat Biotechnol 2022; 40(6): 946-55.
http://dx.doi.org/10.1038/s41587-021-01180-3 PMID: 35145313

[88] Reautschnig P, Wahn N, Wettengel J, *et al.* CLUSTER guide RNAs enable precise and efficient RNA editing with endogenous ADAR enzymes *in vivo*. Nat Biotechnol 2022; 40(5): 759-68.
http://dx.doi.org/10.1038/s41587-021-01105-0 PMID: 34980913

[89] Katrekar D, Yen J, Xiang Y, *et al.* Efficient *in vitro* and *in vivo* RNA editing *via* recruitment of endogenous ADARs using circular guide RNAs. Nat Biotechnol 2022; 40(6): 938-45.
http://dx.doi.org/10.1038/s41587-021-01171-4 PMID: 35145312

[90] Monian P, Shivalila C, Lu G, *et al.* Endogenous ADAR-mediated RNA editing in non-human primates using stereopure chemically modified oligonucleotides. Nat Biotechnol 2022; 40(7): 1093-102.
http://dx.doi.org/10.1038/s41587-022-01225-1 PMID: 35256816

[91] Baker AR, Slack FJ. ADAR1 and its implications in cancer development and treatment. Trends Genet 2022; 38(8): 821-30.
http://dx.doi.org/10.1016/j.tig.2022.03.013 PMID: 35459560

[92] Song J, Dong L, Sun H, *et al.* CRISPR-free, programmable RNA pseudouridylation to suppress premature termination codons. Mol Cell 2023; 83(1): 139-155.e9.
http://dx.doi.org/10.1016/j.molcel.2022.11.011 PMID: 36521489

[93] Zheng Z, Shi N, Xia Q. Adeno-associated viral delivery of engineered tRNA-enzyme pairs into nonsense mutation mouse models. STAR Protoc 2023; 4(1): 101950.
http://dx.doi.org/10.1016/j.xpro.2022.101950 PMID: 36527714

[94] Huang Q, Chan KY, Wu J, *et al.* An AAV capsid reprogrammed to bind human transferrin receptor mediates brain-wide gene delivery. Science 2024; 384(6701): 1220-7.
http://dx.doi.org/10.1126/science.adm8386 PMID: 38753766

[95] Dogbevia G, Grasshoff H, Othman A, Penno A, Schwaninger M. Brain endothelial specific gene therapy improves experimental Sandhoff disease. J Cereb Blood Flow Metab 2020; 40(6): 1338-50.
http://dx.doi.org/10.1177/0271678X19865917 PMID: 31357902

[96] Körbelin J, Dogbevia G, Michelfelder S, *et al.* A brain microvasculature endothelial cell-specific viral vector with the potential to treat neurovascular and neurological diseases. EMBO Mol Med 2016; 8(6): 609-25.
http://dx.doi.org/10.15252/emmm.201506078 PMID: 27137490

[97] Wu G, Liu S, Hagenstein J, *et al.* Adeno-associated virus–based gene therapy treats inflammatory kidney disease in mice. J Clin Invest 2024; 134(17): 174722.
http://dx.doi.org/10.1172/JCI174722 PMID: 39225099

[98] Mei W, Xiang G, Li Y, *et al.* GDF11 protects against endothelial injury and reduces atherosclerotic lesion formation in apolipoprotein E-Null mice. Mol Ther 2016; 24(11): 1926-38.
http://dx.doi.org/10.1038/mt.2016.160 PMID: 27502608

[99] He X, Wang X, Wang H, *et al.* NeuroD1 regulated endothelial gene expression to modulate transduction of AAV-PHP.eB and recovery progress after ischemic stroke. Aging Dis 2023; 15(6): 2632-49.
http://dx.doi.org/10.14336/AD.2023.1213 PMID: 38270116

[100] Zhu W, Shen F, Mao L, *et al.* Soluble FLT1 gene therapy alleviates brain arteriovenous malformation severity. Stroke 2017; 48(5): 1420-3.
http://dx.doi.org/10.1161/STROKEAHA.116.015713 PMID: 28325846

[101] Ravindra Kumar S, Miles TF, Chen X, *et al.* Multiplexed Cre-dependent selection yields systemic AAVs for targeting distinct brain cell types. Nat Methods 2020; 17(5): 541-50.
http://dx.doi.org/10.1038/s41592-020-0799-7 PMID: 32313222

[102] Varadi K, Michelfelder S, Korff T, *et al.* Novel random peptide libraries displayed on AAV serotype 9 for selection of endothelial cell-directed gene transfer vectors. Gene Ther 2012; 19(8): 800-9.
http://dx.doi.org/10.1038/gt.2011.143 PMID: 21956692

[103] Körbelin J, Sieber T, Michelfelder S, *et al.* Pulmonary targeting of adeno-associated viral vectors by next-generation sequencing-guided screening of random capsid displayed peptide libraries. Mol Ther 2016; 24(6): 1050-61.
http://dx.doi.org/10.1038/mt.2016.62 PMID: 27018516

[104] Periasamy R, Patel DD, Boye SL, Boye SE, Lipinski DM. Improving retinal vascular endothelial cell tropism through rational rAAV capsid design. PLoS One 2023; 18(5): 0285370.
http://dx.doi.org/10.1371/journal.pone.0285370 PMID: 37167304

[105] Barbon E, Kawecki C, Marmier S, *et al.* Development of a dual hybrid AAV vector for endothelial-targeted expression of von Willebrand factor. Gene Ther 2023; 30(3-4): 245-54. http://dx.doi.org/10.1038/s41434-020-00218-6 PMID: 33456057

[106] Wu W, Yang Y, Yao F, *et al.* AAV-mediated *in vivo* genome editing in vascular endothelial cells. Methods 2021; 194: 12-7. http://dx.doi.org/10.1016/j.ymeth.2020.12.001 PMID: 33309782

[107] Eriksson E, Moreno R, Milenova I, *et al.* Activation of myeloid and endothelial cells by CD40L gene therapy supports T-cell expansion and migration into the tumor microenvironment. Gene Ther 2017; 24(2): 92-103. http://dx.doi.org/10.1038/gt.2016.80 PMID: 27906162

[108] Rossi C, Lees M, Mehta V, *et al.* Comparison of efficiency and function of vascular endothelial growth factor adenovirus vectors in endothelial cells for gene therapy of placental insufficiency. Hum Gene Ther 2020; 31(21-22): 1190-202. http://dx.doi.org/10.1089/hum.2020.006 PMID: 32988220

[109] Tonetto E, Cucci A, Follenzi A, Bernardi F, Pinotti M, Balestra D. DNA base editing corrects common hemophilia A mutations and restores factor VIII expression in *in vitro* and *ex vivo* models. J Thromb Haemost 2024; 22(8): 2171-83. http://dx.doi.org/10.1016/j.jtha.2024.04.020 PMID: 38718928

[110] Simmons AB, Bretz CA, Wang H, *et al.* Gene therapy knockdown of VEGFR2 in retinal endothelial cells to treat retinopathy. Angiogenesis 2018; 21(4): 751-64. http://dx.doi.org/10.1007/s10456-018-9618-5 PMID: 29730824

[111] Milani M, Canepari C, Assanelli S, *et al.* GP64-pseudotyped lentiviral vectors target liver endothelial cells and correct hemophilia A mice. EMBO Mol Med 2024; 16(6): 1427-50. http://dx.doi.org/10.1038/s44321-024-00072-8 PMID: 38684862

[112] Merlin S, Cannizzo ES, Borroni E, *et al.* A novel platform for immune tolerance induction in hemophilia a mice. Mol Ther 2017; 25(8): 1815-30. http://dx.doi.org/10.1016/j.ymthe.2017.04.029 PMID: 28552407

[113] Ryu JY, Cerecedo-Lopez C, Yang H, Ryu I, Du R. Brain-targeted intranasal delivery of protein-based gene therapy for treatment of ischemic stroke. Theranostics 2024; 14(12): 4773-86. http://dx.doi.org/10.7150/thno.98088 PMID: 39239521

[114] Feng Y, Guo M, Liu W, *et al.* Co-self-assembly of cationic microparticles to deliver pEGFP-ZNF580 for promoting the transfection and migration of endothelial cells. Int J Nanomedicine 2016; 12: 137-49. http://dx.doi.org/10.2147/IJN.S107593 PMID: 28053529

[115] Hao X, Li Q, Lv J, *et al.* CREDVW-linked polymeric micelles as a targeting gene transfer vector for selective transfection and proliferation of endothelial cells. ACS Appl Mater Interfaces 2015; 7(22): 12128-40. http://dx.doi.org/10.1021/acsami.5b02399 PMID: 26011845

[116] Zhang X, Jin H, Huang X, *et al.* Robust genome editing in adult vascular endothelium by nanoparticle delivery of CRISPR-Cas9 plasmid DNA. Cell Rep 2022; 38(1): 110196. http://dx.doi.org/10.1016/j.celrep.2021.110196 PMID: 34986352

[117] Bozoglu T, Lee S, Ziegler T, *et al.* Endothelial retargeting of AAV9 *in vivo*. Adv Sci 2022; 9(7): 2103867. http://dx.doi.org/10.1002/advs.202103867 PMID: 35023328

[118] Tian S, Feng M, Cao D, *et al.* Endothelial cell-targeted pVEGF165 polyplex plays a pivotal role in inhibiting intimal thickening after vascular injury. Int J Nanomedicine 2015; 10: 5751-68. http://dx.doi.org/10.2147/IJN.S88109 PMID: 26425083